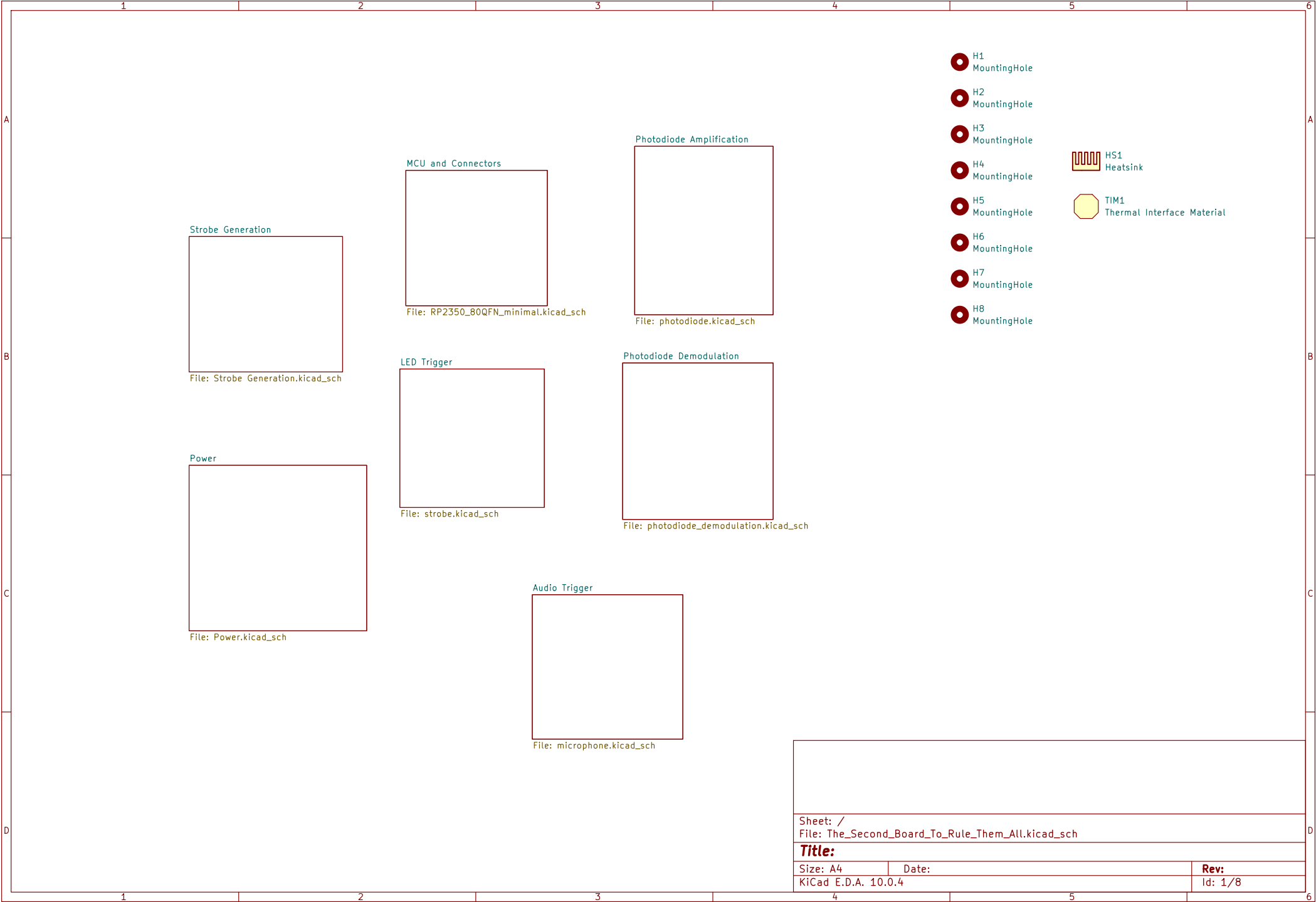
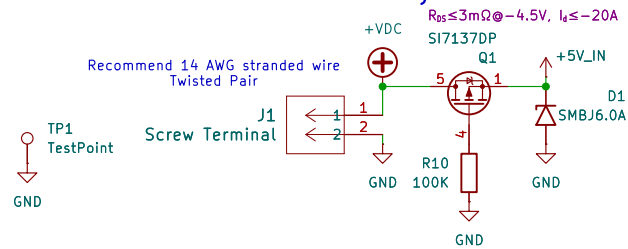


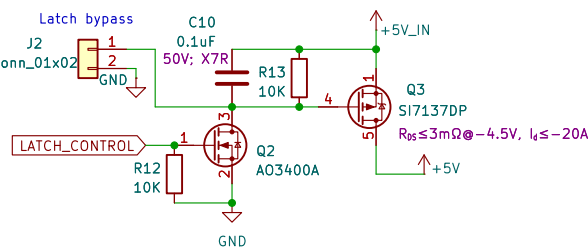


Power in must be 5.2V

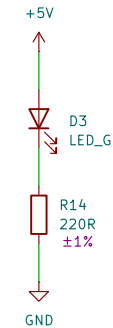
Reverse Polarity Protection



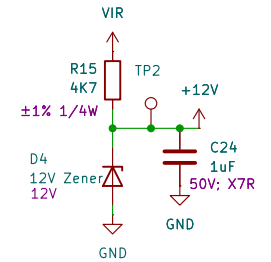
Power Latch



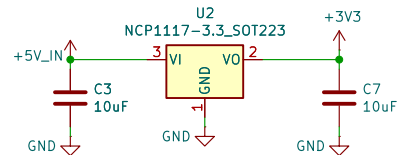
Power LED



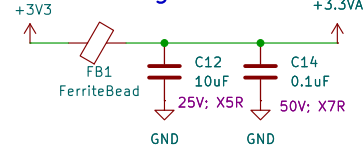
12V Rail



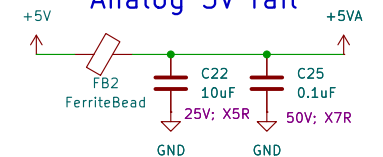
3V3 Rail



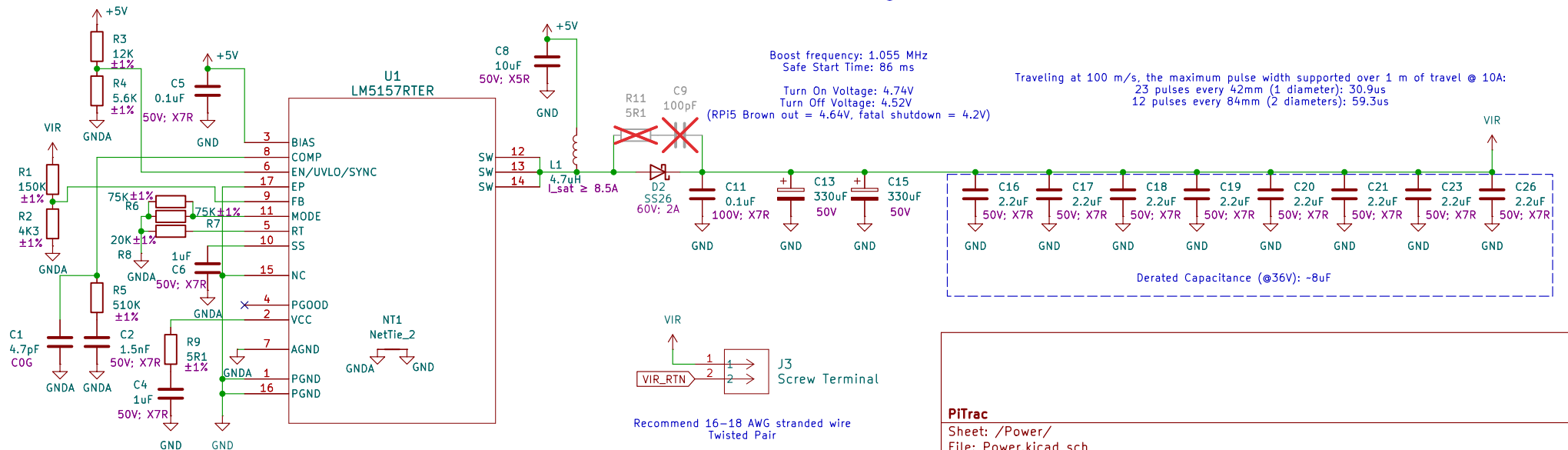
Analog 3V3 rail



Analog 5V rail



36V Boost Converter @ 0.4A



PITrac

Sheet: /Power/
File: Power.kicad_sch

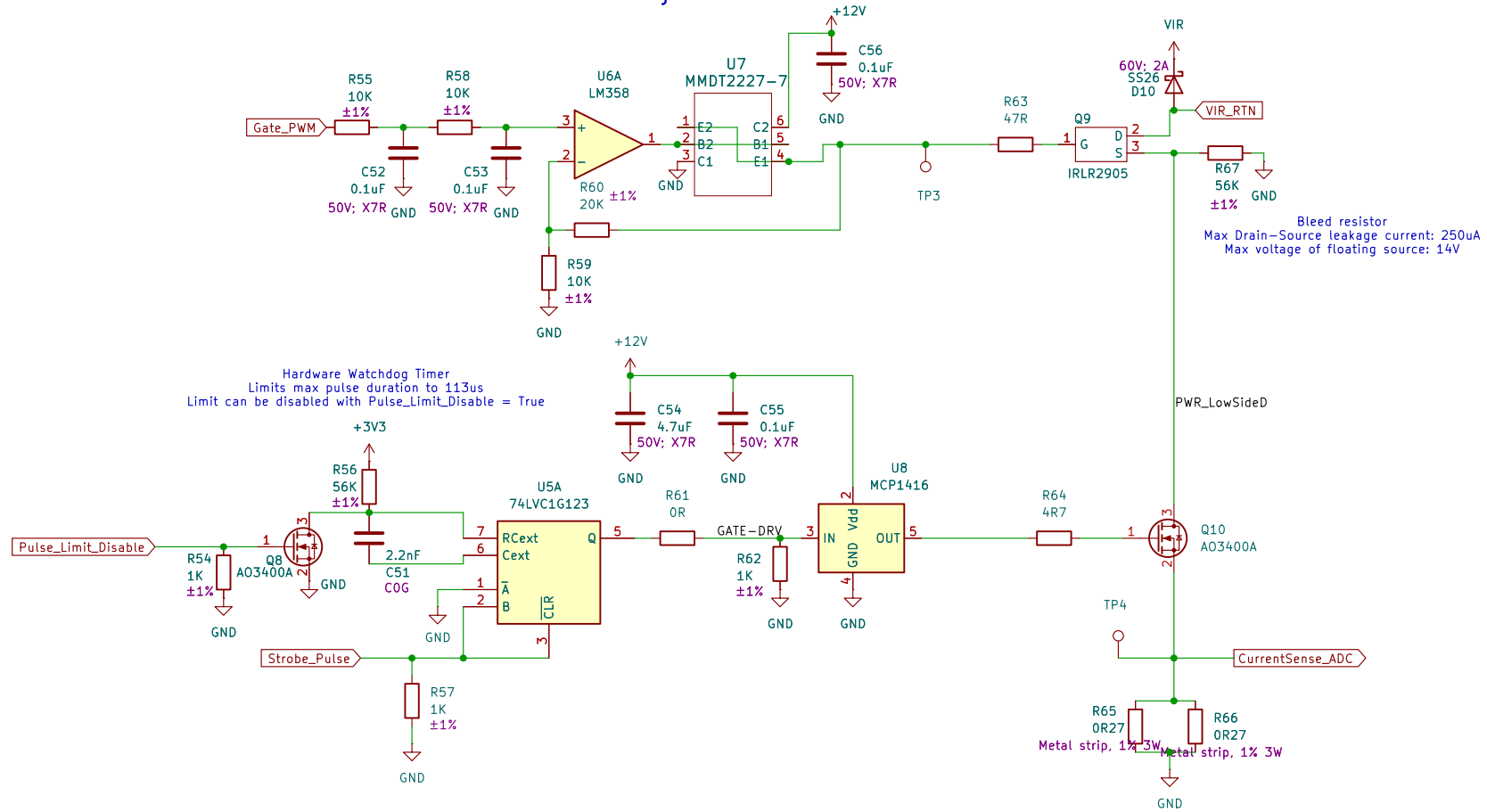
Title: V3 Connector + IRLED

Size: A4 Date: 2026-02-12
KiCad E.D.A. 10.0.4

Rev: V1
Id: 2/8



Adjustable Constant Current Sink



PiTrac

Sheet: /Strobe Generation/
File: Strobe Generation.kicad_sch

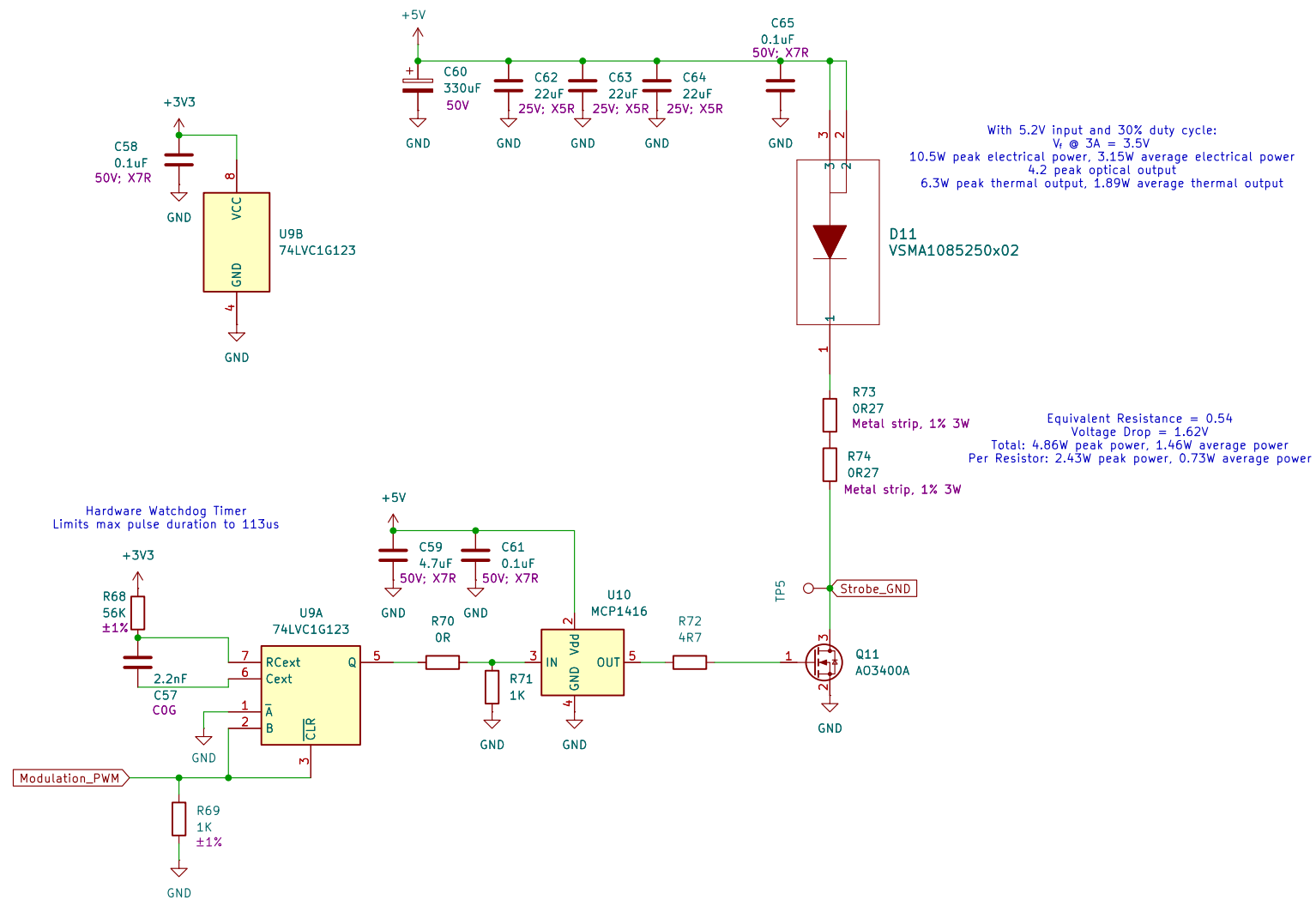
Title: V3 Connector + IRLED

Size: A4 Date: 2026-02-12

KiCad E.D.A. 10.0.4

Rev: V1

Id: 4/8



30% Duty Cycle

Sheet: /LED Trigger/
 File: strobe.kicad_sch

Title:

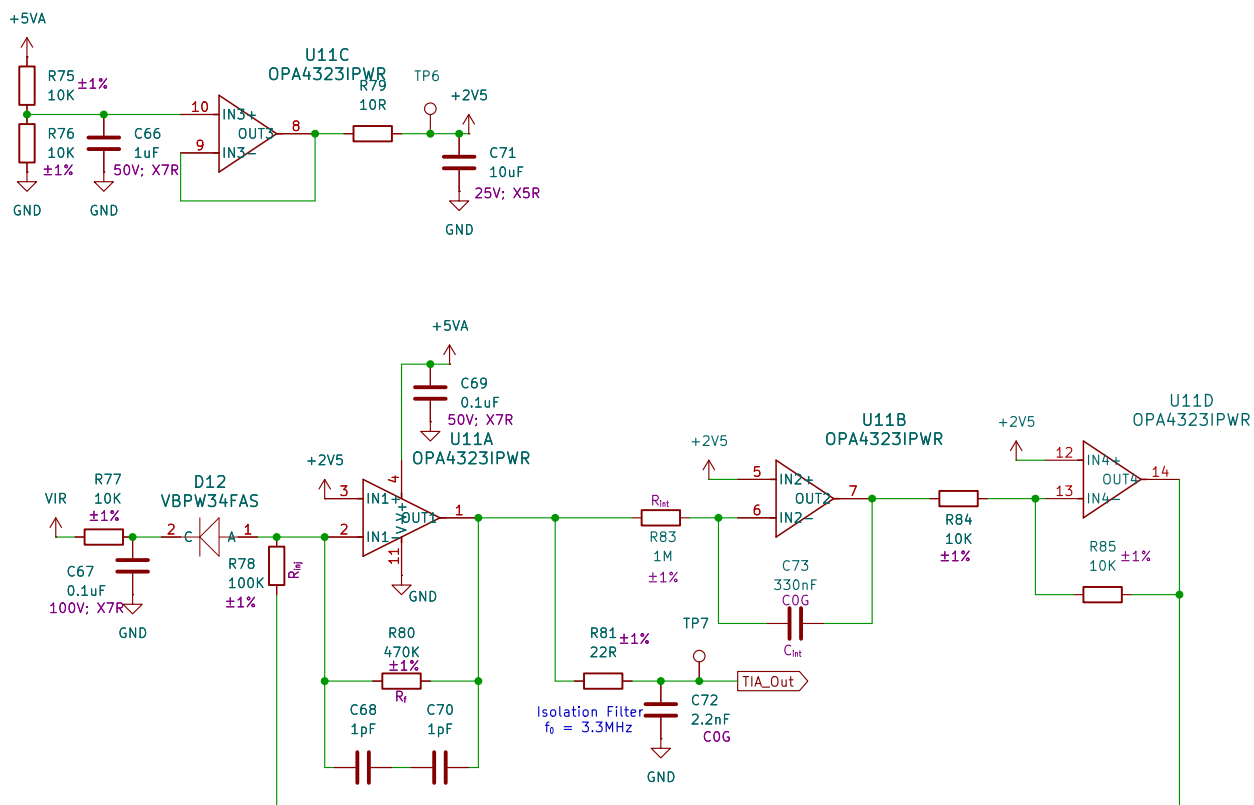
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Date:

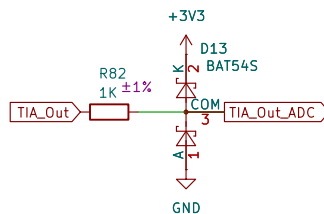
KiCad E.D.A. 10.0.4

Rev:

Id: 5/8



Transimpedance Amplifier TIA with DC Servo
 Gain = $-R_f = -4.7e5$
 $f_0 = R_f / (2 * \pi * R_{in} * R_{int} * C_{int}) = 2.27Hz$



Sheet: /Photodiode Amplification/
 File: photodiode.kicad_sch

Title:

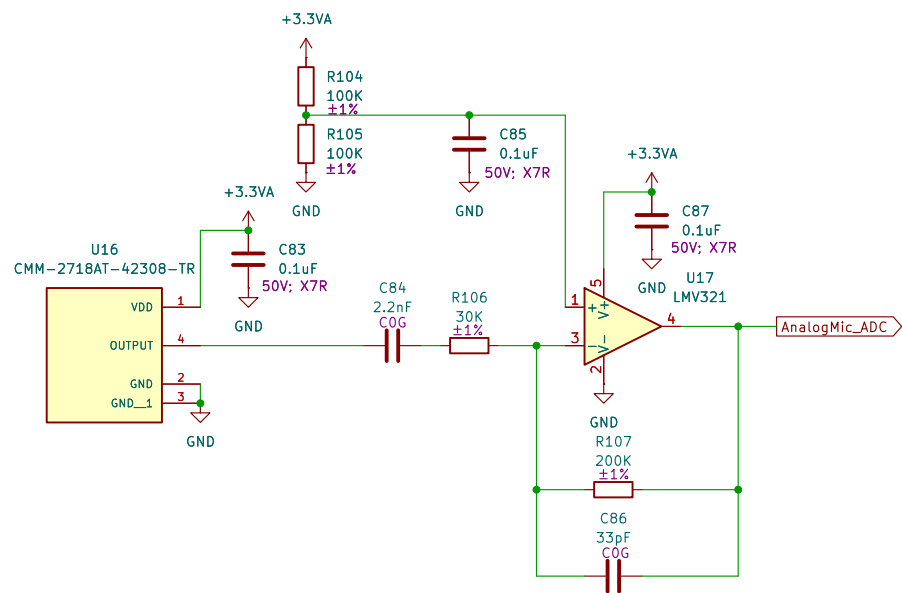
Size: A4

Date:

KiCad E.D.A. 10.0.4

Rev:

Id: 6/8



Sheet: /Audio Trigger/
File: microphone.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.4

Rev:

Id: 8/8